Scale, Asymmetry, and the Structure of Distinction
An Essay (Full Version)

Serge Magomet (aka Aimate)

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"The arithmetic we use to count apples and sheep has become a time bomb now that we apply it to the quantum world and the cosmos."

Abstract

This essay argues that the foundational crisis in contemporary physics and mathematics stems not from insufficient data, but from the uncritical application of a scale-local language—classical arithmetic—to ontologies operating at radically different regimes of distinction. We propose a methodological shift: treating *scale* not as an external metric, but as an ontological operator that structures the very possibility of objects, numbers, space, and the observing "I."Developing a framework of *operational phenomenology*, we identify a fundamental asymmetry along the scale axis: a rigid lower limit (where distinction dissolves into fluctuation) and an unbounded upper horizon (where organization persists). This asymmetry is shown to be a precondition for any act of distinction, not a derivative physical property. Within this frame, we reinterpret key cosmological phenomena—dark energy, the cosmological horizon, the Big Bang—not as physical substances or events, but as effects of transitions between regimes of distinction.

This extended version complements a previously released concise summary. While the short version outlines the core thesis and findings, this full text is indispensable because it preserves the "texture"of the inquiry—the living fabric in which concepts emerged from immediate phenomena rather than from abstract definitions. It is particularly valuable not, but for documenting the methodological process and the emergence of insights themselves. By tracing the genealogy of these ideas, the extended version offers a transparent view of the analytical journey, allowing readers to engage with the operational phenomenology method in practice. This makes the work a resource not only for foundational theory but for understanding how ontological shifts are discovered and articulated.

Keywords: ontology of scale, operational phenomenology, foundations of mathematics, asymmetry, distinction, quantum gravity, cosmology, category theory

Here is a simple question that might seem naive even to a schoolchild: where exactly does "arithmetic" cease to apply?

At first glance, the answer seems obvious. Arithmetic works everywhere—from counting apples to calculating cosmic trajectories. What could be more reliable than $1 + 1 = 2$? Mathematical operations are symmetric, reversible universal.

Yet it is precisely this obviousness that proves to be a trap.

Upon deeper investigation, a strangeness emerges. We know that there is uncertainty, randomness, probability—things that clearly do not fit within the framework of classical arithmetic. Somewhere between counting apples and quantum mechanics, a rupture occurs. But where exactly?

The next step leads to even deeper perplexity. The number π, the circle, and the square belong to different ontologies. The square lives in the world of linear equations; the circle, in the world of transcendental functions. The attempt to "square the circle" is not a technical problem but a collision of two languages for describing reality.

Stop!—one might say.

> *The arithmetic we use for apples and sheep became a time bomb when we began applying it to the quantum world and the cosmos.*

This is not a metaphor. This is a diagnosis.

What is happening is that we continue to employ the same basic language—the same operations, the same numbers—to describe radically different ontologies. Yet we do not rethink arithmetic itself. We simply layer new tools atop the old, creating an illusion of progress.

This holds even when dealing with large numbers, infinities, or infinitesimals.

It is then that we recall the concept of scale.

Scale is not merely "size" or "unit of measurement." It is a parameter that determines the very structure of distinction.

Introduction: The Character and Methodology of the Inquiry

Let us attempt to rethink the concept of scale, setting aside familiar notions (size, unit of measurement, counting). Instead, scale is treated as a parameter that determines the structure of the very act of distinction—that which enables the appearance of objects, numbers, space, and even the "I" itself.

Key methodological stance: rejecting binary oppositions (external/internal, homogeneous/inhomogeneous, micro/macro) and the "nesting-doll" hierarchy of worlds in favor of their juxtaposition. The investigation proceeds within the tradition of operational phenomenology.

Below, we outline the ideas we find most original and promising. They are grouped to preserve the "texture" of our reflections—the living fabric in which concepts emerged from immediate phenomena rather than from abstract definitions.

Scale as an Operator of Distinction, Not a Measure

The Initial Intuition

Traditional mathematics and physics treat scale as an external parameter (a unit of measurement) that can be applied to the world while remaining

unchanged. The dialogue proposes a different view: scale is an operation that constitutes the very possibility of distinction.

Instead of asking "how large is the object?", let us begin with the question: "what happens when I change the degree of approximation?"

It turns out that what changes is not merely the "picture," but the very structure of what object. At one scale, the field of view is smooth; at another, folds and irregularities appear within it that can be singled out as separate formations; at a third, these formations dissolve into something that can no longer be described in terms of the original objects.

Clarification: "Dissolution into noise" vs. Change of Ontology

It is important to clarify that "dissolution" does not mean a transformation into structureless chaos. In physics, noise often carries information (the fluctuation-dissipation theorem, quantum fluctuations, correlation functions). Our position is that when decreasing scale, we encounter not simply "noise," but a fundamental change in the regime of distinction. This means:

- *Practical limit*—the limitation of a given perceptual channel (resolution of the eye, sensor pixel, thermal noise in an amplifier). Beyond this, there is still structure, but it is inaccessible through this channel. This is not an ontological boundary.

- *Ontological limit*—the scale at which the concepts forming the basis of the given field of distinction (space, time, object, identity) cease to function. This is not "finer detail," but a change of ontology. In physics, an example is the Planck scale, where the metric and topology of spacetime lose their usual meaning, and structures emerge that are described by spin networks, loop quantum gravity, noncommutative geometry.

Argument Against Hidden Variables

Hidden variables assume that behind probabilistic behavior there is a deterministic but inaccessible structure, expressed in the same terms (space, time, trajectories). Our position is different: even if such a structure exists, it cannot be expressed in the same terms as on the original scale. This is not a matter of incomplete knowledge, but of incompatible ontologies. The operational argument: when approaching the Planck scale, the operator of distinction ceases to be defined as a mapping that preserves the basic categories (for example, the metric). A hidden variable defined in terms of the old ontology will be unable to describe the new ontology without committing a category error.

Thus, the asymmetry of scale consists not in the fact that below there is noise, but that below we encounter a fundamental change above (when increasing scale) new levels can be described as a complication of the same forms (hierarchy, emergence), without a change in the basic categories.

Operational Grounding

Before proceeding to formalization, let us register what is given to us in immediate experience.

The experience of consistently changing the scale of attention reveals:

1. The field of distinction exists prior to the isolation of objects. At one scale, the field appears homogeneous; at another, structures emerge within it. This accords with the earlier definition of form as a wholeness preceding dissolution into objects and background.

2. The order of scale change matters—zooming in and zooming out are asymmetrical. Zooming in, we sooner or later encounter a limit; zooming out, no such limit exists.

3. In one direction ("inward"), there is a rigid limit of distinction—the structure disintegrates into fluctuations, instability. In the other direction ("outward"), there is no limit—the object simply leaves the field of view, yet the field remains distinguishable.

The Tripartite Structure of Distinction: Field, Point, Boundary

In immediate experience, we discover a tripartite structure: the field of distinction, the point of reference ("here"), and the boundary that makes this distinction possible while remaining invisible from within.

When we "change scale," we do not merely zoom in or out on a picture. We alter the entire triad:

- The field of distinction ceases to be a homogeneous background—new inhomogeneities appear within it, or, conversely, previous structures "dissolve."

- The point of reference shifts—what was an "object" becomes "background," and vice versa.

- The boundary is redrawn—what was "inside" perception recedes beyond the horizon, while new elements intrude from "outside."

To avoid anthropocentrism, we adopt the following approach: the tripartite structure—field, point of reference, boundary—is first registered in self-observation. It is then found to be reproduced in other systems. The tablet through which the dialogue with AI is conducted is for me an external object, but for the AI it is part of its own "body." Colliding billiard balls do not "see" each other's interiors; their interaction entails a "blind spot" (mass distributed throughout the volume invariants. A quantum particle is subject to a fundamental limit on the simultaneous measurement of position and momentum—this is not an instrumental limitation but a property of reality itself.

Fractals and Anti-Fractals

For a long time, fractals were regarded as a universal model of reality's structure. Self-similarity at all scales promised symmetry and harmony: look closer, and you see the same thing; move farther away, and again the same thing.

But reality proves otherwise.

In the microworld, fractal logic breaks down. We encounter a threshold beyond which "looking closer" loses its meaning—the uncertainty principle, quantization, possibly the discreteness of spacetime. The world at small scales is not fractal in nature. It does not preserve self-similarity.

If a fractal is a structure that repeats itself with a change of scale microworld we observe something else: a structure that qualitatively changes when transitioning to another scale. Discrete networks, probabilistic clouds, nonlocal correlations, topological defects—none preserve self-similarity under scaling.

This is the first indication of a fundamental asymmetry along the scale axis: in one direction, the potential for infinite repetition; in the other, a rigid limit.

Asymmetry of the Scale Axis and Its Ontological Status

Key Thought

Asymmetry (a rigid limit when moving "inward," no limit when moving "outward") is not a property of the world explicable by physics. It is a condition for the possibility of any distinction whatsoever. Without it, there would be neither objects nor an observer.

History of the Insight

We repeatedly returned to the question: why can we mentally "expand" to the size of a galaxy, but cannot "shrink" to the size of an atom while preserving the integrity of perception? Why can consciousness not be arbitrarily small? The answer we arrived at: because the structure of distinction has a threshold from below. Below this threshold, distinction disintegrates (in the sense of a change of ontology, not the appearance of "fine noise"), and there is nothing that could be called an "I" or an "object" in familiar terms. Above, however, there is no such threshold—one can conceive of the "I" of a planet, a galaxy, the universe, and even if the universe is finite, the very possibility of distinction does not disappear.

This conclusion was supported by examples from everyday life: family, tribe, state, global society—structures that scale upward without apparent limit; below, there is a limit—one cannot have a "state of two people" with full institutions. In biology: an organism can be huge (whale, sequoia), but cannot be arbitrarily small and remain complex. In technology: a computer can be made larger (servers), but cannot be made arbitrarily small while retaining power.

Why Asymmetry Is Not an Epistemological Limitation

A possible objection: the asymmetry of scale may not be a property of the world itself limitation of our cognitive apparatus—that we cannot see "too small" and cannot encompass "too large" due to the structure of our sense organs and brain. Then our ontology would be a form of transcendental idealism, where properties of reality are derived from the conditions of perception. The response to this objection rests on four arguments.

1. Asymmetry manifests not only in human perception

The described structure is found in systems where there are neither sense organs nor consciousness. Two colliding billiard balls do not "see" each other's interiors; their interaction has a "blind spot" (mass distributed throughout the volume) that participates through invariants. A quantum particle has a fundamental limit on the simultaneous measurement of position and momentum—this is not a limitation of the instrument but a property of reality itself. The cosmological horizon is not something we cannot see because of the finite speed of light, but a boundary beyond which the distinction between "there" and "here" loses

meaning. Asymmetry is reproduced at all levels, from physics to biology, without reference to human perception.

2. Asymmetry cannot be eliminated by improving instruments

If asymmetry were merely epistemological, we might expect that with technological development (more powerful microscopes, telescopes, detectors) it would shift and eventually disappear. However, in physics we encounter fundamental limits: the Planck length cannot be "overcome" by any instrument, because at that scale the very concept of distance loses meaning. Similarly, the cosmological horizon cannot be "overcome" even in principle, because beyond it there is no spacetime in our understanding. These boundaries are not practical, but ontological.

3. The "I" is not the source of asymmetry but one of its particular manifestations

The "I" is not the source of asymmetry but one of its particular manifestations. It is not that the world is symmetric and perception asymmetric; rather, perception is asymmetric because the world is structured that way.

4. Ontological status as a condition of possibility

Asymmetry of scale is neither an empirical fact (which could be refuted by a future experiment) nor a purely epistemological limitation. It is a condition for the possibility of distinction itself: if there were no asymmetry, there would be no stable objects, no observers, and no question of cognition. This makes it an ontological principle that cannot be "removed" by better knowledge, but can be discovered in any act of distinction.

Asymmetry as a Primary Ontological Principle

Our position is *operational realism*: the asymmetry of scale is a property of reality that manifests in all systems and is accessible to any act of distinction. The problem of quantum gravity thus appears not as the search for a unified theory, but as the study of transitions between regimes of distinction.

Form and Geometry Are Prior to Arithmetic (On the Question of "New Mathematics")

The Initial Intuition

We repeatedly returned to the idea that arithmetic (counting, numbers, operations of addition) arises only within a certain range of scales, where the field of distinction stabilizes into stable, isolated objects. What is primary is not arithmetic, but form and its transformation with a change of scale.

What is form?

Form is a wholeness given prior to any distinction into parts. It is not a geometric figure, but rather something that has not yet dissolved into objects and background. Form changes with a change of scale, but this is a change of the whole, not merely the appearance of parts.

When we "change scale," we do not single out objects. We alter the form of the field itself:

- At one scale, the field appears smooth and continuous.
- At another, folds, curves, and irregularities appear within it.
- At a third, these irregularities acquire their own structure, their own geometry.

Form changes. But this is not "the appearance of objects"—it is a change of the whole.

Asymmetry of form

When decreasing scale, form does not simply become more detailed—it disintegrates: a smooth surface turns into noise, into chaos, into something that can no longer be called "form."

When "increasing scale," form does not disintegrate—it organizes. New levels of structure emerge, yet wholeness is preserved.

This is an asymmetry not of counting, but of form.

Where is the boundary here?

The boundary is not a line between objects. It is the limit of form. The point beyond which form ceases to be form.

And this boundary is asymmetrical: in one direction, disintegration; in the other, organization.

What does "new mathematics" mean?

If we abandon counting as primary, then mathematics must begin with the act of distinguishing form. This does not mean abandoning numbers, but it requires rethinking their status. Numbers become not "ideal objects," but invariants that arise in that mode of the field of distinction where (as shown in the section on countability) objects acquire the stability and isolation that allows them to be compared as "units." Outside this mode, numbers either do not work or work differently.

What such a mathematics might look like:

- *Basic concepts:* not "set" and "element," but field of distinction and scale operator. The field is that which is given prior to division into objects. The scale operator is a way of changing the depth of distinction, observing how the structure of the field changes.

- *Operations:* not addition and multiplication, but extraction (transition from homogeneous to inhomogeneous), merging (the reverse transition), comparison (establishing relations between inhomogeneities), transition (changing scale).

- *Invariants:* not numbers, but topological properties of form that are preserved with a change of scale (connectivity, number of holes, type of singularity).

- When stable, isolated inhomogeneities arise in the field, the possibility of counting them emerges. But this is a secondary, local regime. Arithmetic turns out to be a special case, a dialect that operates within a certain range of scales.

Caution Regarding "New Mathematics"

The phrase "new mathematics" might create the false impression that we are proposing a ready-made alternative formal system competing with existing ones. This is not the case. Our goal is more modest, but perhaps deeper.

We propose not a *new mathematics*, but a *meta-language for the foundation of mathematics*—a phenomenology of mathematical experience. Our task is not

to construct an axiomatic from which addition and multiplication could be derived (though in the long term this is possible), but to understand:

- Why is arithmetic possible at all?
- Why does it work only within a certain range of scales?
- Why does it "break" outside this range (the quantum world, cosmology)?

Thus, the proposed approach is an *investigation into the conditions of possibility of mathematics*, not its replacement. The proposed axioms (field, operator $\Gamma(\lambda)$, asymmetry) and the sketch of a graph algebra are not an alternative mathematics, but rather a *conceptual framework* within which existing mathematics can be understood as a particular, scale-limited regime.

In the long term, this framework could be formalized within category theory or topos theory, which would allow speaking of *different mathematics* for different scales. But at this stage, we remain at the level of problem formulation and a sketch of a possible path, avoiding premature formalization that might obscure the initial intuition.

Uniqueness of the idea

This is not simply "geometry is more important than arithmetic," but a radical shift: arithmetic is not a universal language, but a local dialect that arises only within a certain range of scales. This explains why mathematics "break" at the boundaries (the quantum world, cosmology): there, the conditions for the emergence of arithmetic operations are simply absent.

When Does Countability Emerge?

Countability ($1 + 1 = 2$) presupposes stable, isolated objects. However, objects arise only when the boundary acquires a certain structure—when the limit becomes "smooth" and reproducible within a certain range of focal lengths.

Outside this range, there are no objects, no counting. Not because "the laws are different," but because the conditions for the very phenomenon of the object to arise are absent.

Arithmetic is not a universal language of reality. It is a local dialect, operative only in that mode of attention where the boundary stabilizes in a certain configuration.

Geometry (form, topology, structure) is primary: it describes how the field of distinction is organized prior to the emergence of countable objects within it. A change of scale is a transformation of form, not a change in the number of objects.

Resonance with Contemporary Mathematical Programs: Categorical Logic and Topos Theory

The idea of the primacy of form (geometry, structure of relations) over arithmetic (counting, numbers) is not an isolated intuition. It finds its development in contemporary approaches to the foundations of mathematics that seek to restructure mathematical knowledge starting not from set theory and discrete objects, but from the concepts of morphism, category, and topos.

In categorical logic, objects are defined not through their internal structure (as a "set of elements"), but through relations (morphisms) to other objects. This

shift of focus from "what is" to "how it is connected" directly corresponds to our operational stance: the primary is the act of distinction, which in categorical language can be expressed as an arrow connecting objects in a network of relations. Counting and numbers emerge here as a special case—as a description of certain types of structures (for example, natural numbers as the natural number object in the category of sets).

Topos theory goes further, offering a universal language in which different "worlds" (topoi) with different internal logics can be constructed. In a topos, the concept of number can be defined, but it will depend on the internal structure of the topos, not on an absolute, predetermined arithmetic. This resonates with our thesis: arithmetic is not universal, but is a local, scale-dependent regime of the field of distinction. Different topoi are analogous to different regimes of distinction, each with its own internal logic.

Moreover, in topos theory, the concept of "truth value" (subobject classifier) arises from the structure of the topos itself, not imported from outside. This resembles our idea that invariants (numbers, logical values) appear as stable patterns in the field of distinction, not as absolute beginnings.

Comparison with Alternative Programs

The proposed framework is not isolated. Below, we briefly compare it with three influential programs to clarify its specificity and avoid misunderstandings.

1. In Comparison with Relational Quantum Mechanics (Rovelli, Laudan)

Relational quantum mechanics asserts that the state of a system exists only relative to another system. Our ontology agrees that the act of distinction (interaction) is primary, and objects are secondary. However, relational quantum mechanics remains within physics and does not raise the question of the structure of distinction itself. We go further, attempting to derive from scale asymmetry not only quantum effects, but also the foundations of mathematics and the phenomenology of the "I."

2. In Comparison with William Lawvere's Program (Elementary Theory of the Category of Sets, ETCS)

Lawvere and his followers proposed a foundation for mathematics based on category theory rather than naive set theory. This resonates with our rejection of "set" as a primary concept. However, in ETCS, objects (sets) are defined through relations (morphisms), but the question of why these relations are possible at all and why they have an asymmetric structure is not raised. Our framework attempts to answer this question by introducing the scale operator and asymmetry as an ontological primitive.

3. In Comparison with Constructivism (Brouwer's Intuitionism, Martin-Löf Type Theory)

Constructivism asserts that mathematical objects are constructed, not discovered. Our work agrees with this in its critique of Platonism. However, constructivism often focuses on construction procedures within a formal system, without explaining why these procedures have the structure they do. Our approach, by contrast, attempts to derive the structure of mathematical operations (including counting and arithmetic) from the ontology of distinction. We are not building an alternative formal system, but proposing a meta-language for understanding why formal

systems are possible at all and why they have the structure they have.

Conclusion: Our work does not directly compete with these programs, but offers for them an *ontological grounding*—an explanation of why the structure of mathematics and physics is asymmetric, scale-dependent, and rooted in the act of distinction.

By pointing out this connection, we position our work not as an isolated philosophical reflection, but as resonating with contemporary trends in the foundations of mathematics, where "naive set theory" is being replaced by structural, relational, ontologically flexible languages. Categorical logic and topos theory provide a powerful formal apparatus for realizing what we called "formal phenomenology of scale"—a language not tied to numbers, but allowing the description of transformations of forms. In this sense, our investigation offers not an alternative to these approaches, but their phenomenological grounding and extension to the problem of scale.

The "I" as a Prototype of the Field of Distinction and an Operator of Scale

The Initial Intuition

We advance the idea that the phenomenon of the "I" (pure presence, the point of reference) serves as the basic model of the field of distinction. Its structure—the boundary between "I" and "not-I," the absence of spatial localization of this boundary, the asymmetry of scale—is reproduced at all levels of reality.

Let us attempt to hold the "I" as pure experience, without attributing to it either form or size. Then it turns out that the first thing that emerges from this experience is a limit. Not a line on a map, but a sensation: "further—cannot." This limit has no fixed location; it is everywhere and nowhere, like the center of a sphere that coincides with every point on its surface. When the scale of attention (focus) changes, it is not the size of the "I" that changes, but the structure of this limit: it can be rigid, blurred, generating new levels or disintegrating into noise (in the sense of a change of ontology).

The "I" as Singularity of Distinction

The phenomenon of the "I" (pure presence, the point of reference) serves as the basic model of the field of distinction. Its structure—the boundary between "I" and "not-I," the absence of spatial localization of this boundary, the asymmetry of scale—is reproduced at all levels of reality.

In experience, the first thing that emerges from the pure experience of the "I" is a limit—not a line on a map, but a sensation: "further—cannot." This limit has no fixed location; it is everywhere and nowhere, like the center of a sphere that coincides with every point on its surface.

The image of the "I" as a wormhole (cf. Leibniz's monads) arises from the following observation: the "I" is a point through which infinite worlds pass, yet the point itself belongs to none of them. The monad—a mathematical point behind which lies infinity—here acquires operational meaning: the "I" is such a point, and upon a change of scale, it can unfold into entire worlds or collapse into pure being.

The Asymmetry of the "I"

The asymmetry of the "I" is analogous to that of the scale axis. Below (the extremely small), there is a rigid limit: an atom does not possess an "I"; an

electron does not reflect. There is a minimum scale below which the "I" dissolves into non-distinction.

Above (expansion), there is no such limit: one can imagine the "I" of a planet, a galaxy, the universe. The limit here is practical rather than ontological.

This asymmetry is not derived from other principles; it is given as a condition for the possibility of distinction itself. Without it, there would be neither objects nor an observer.

Connection with Leibniz's Monads and Wormholes

An image emerges of human "I"s as wormholes—points through which infinite worlds pass, yet the point itself belongs to none of them. This unites asymmetry (collapse below, expansion above), the boundary (having no location), and the idea that the "I" is a condition for the possibility of distinction, not an object within it. Leibniz's monad—a mathematical point behind which lies infinity—here acquires not a metaphysical but an operational meaning: the "I" is such a point, and upon a change of scale, it can unfold into entire worlds or collapse into pure being.

Uniqueness of the Idea

The "I" appears not as a special case of consciousness, but as an ontological prototype of any distinction. This allows us to rethink scale: scale is not an external parameter, but a parameter of the depth of the "I"—a parameter of how far the act of distinction can extend without destroying itself. This provides a natural bridge between phenomenology and ontology.

The Dual Meaning of "I": Phenomenological Point of Reference and Universal Principle of Distinction

To avoid accusations of anthropocentrism, it is necessary to clearly distinguish two senses of "I," which in the dialogue were often conflated but must be explicitly separated.

First sense: "I" as phenomenological givenness

In this sense, the "I" is that which is directly given in experience: the point from which perception, experience, and reflection are conducted. This is not an essence or an object, but an indexical sign without which the very act of givenness would not exist. It is in this sense that we spoke of "glancing around," of the boundary between "I" and "not-I," of the impossibility of seeing the back of one's own head. This sense is tied to a specific observer (a human, possibly an AI) and cannot be directly transferred to other systems.

Second sense: "I" as universal principle of distinction

In this deeper sense, the "I" is not a human being or a subjective center, but a structural position that arises wherever an act of distinction occurs. This could be termed the "point of reference"—that relative to which the field of distinction is structured, divided into "here" and "there," into "self" and "other." This principle is not anthropomorphic: it manifests in quantum interaction (where one particle "distinguishes" itself from another through correlations), in the operation of a sensor (where a signal is distinguished from noise), and in social structures (where a group distinguishes itself from other groups).

In this second sense, the "I" is not a property of humans, but an ontological invariant: any system capable of distinction necessarily contains within it a

structural analogue of a point of reference, a boundary, and a field of the distinguished. This is not panpsychism (we do not ascribe "consciousness" to a stone or an electron), but rather the claim that the structure of distinction, which in humans is experienced as the "I," can be described in other systems in terms of relations that do not presuppose subjectivity.

How to Avoid Anthropocentrism

We do not "project" the human I onto the world. On the contrary, we use human experience as empirical access to a structure subsequently discovered to be universal. The methodological move is as follows:

1. In self-observation, we register the tripartite structure: field, point of reference, boundary.

2. We then show that this structure is reproduced in other systems (billiard balls, sensors, mathematical theories)—not by analogy, but by virtue of their having a "blind spot" (that which is not given in the act of interaction but constitutes it).

3. From this, we conclude that we are dealing not with a human peculiarity, but with an ontological invariant that can be described without reference to human consciousness.

Thus, the "I" in the second sense is not a metaphysical entity, but an operator introduced to describe any act of distinction. It could be called the "point of reference," the "act of extraction," the "primordial singularity"—but the essence remains the same: without it, there is no structure, no "here" and "there," no objects.

Connection with the Leibnizian Monad

The image of Leibniz's monad is apt here not as a metaphysical doctrine, but as a precise indication that each monad (any system capable of distinction) reflects the entire world, but from its own unique point of reference. This does not mean that an atom has a "soul"; it means that in any interaction, one can identify a structural position analogous to a point of reference. In this sense, the "I" as universal principle is not anthropocentrism, but a generalization of phenomenological structure to any ontology.

The Status of the "I": Between Realism and Idealism (The Field of Distinction as Ontological Primitive)

A possible provocative question—"Is the 'I' a necessary condition for the existence of the field of distinction in the universe before the appearance of observers (realism), or does the field of distinction arise only together with the observer (idealism)?"—forces a choice between two classical positions. However, our ontology, grounded in the MPO-System and operational phenomenology, does not require such a choice, because the dichotomy "before the appearance of the observer" and "together with the observer" itself turns out to be secondary to the act of distinction.

In the MPO-System, there is no priority of one over the other. The "I" (as point of reference) and the field of distinction arise simultaneously in the act of

distinction. This does not mean they are "created" by the observer, nor does it mean they exist independently. They are two aspects of a single event—an act in which something is distinguished from the background and, simultaneously, the position from which this extraction is performed is fixed.

Methodological Clarification: Phenomenological Givenness and Ontological Hypothesis

Before proceeding further, it is necessary to explicate the methodological status of our transition from the "I" as phenomenological givenness to the "I" as universal principle of distinction. This is not the result of phenomenological reduction in the strict Husserlian sense, but a conscious adoption of a metaphysical hypothesis—*operational realism*.

The phenomenological method allows us to register the tripartite structure (field, point of reference, boundary) in the experience of self-observation. However, the claim that this same structure is reproduced in quantum interactions, billiard balls, or social systems is *not derived* from phenomenological analysis. It is an ontological extrapolation adopted as a working hypothesis.

We do this not arbitrarily, but guided by the principle of *operational symmetry*: if structurally analogous relations can be discovered in different systems (human, sensor, quantum particle)—the presence of a "blind spot," irreversibility of interaction, asymmetry of limits—then we are entitled to suppose that we are dealing with an ontological invariant, not an artifact of consciousness.

Thus, our position is a hybrid:

- *Phenomenological basis*: the initial data are taken from the experience of self-observation.
- *Realist hypothesis*: the structure of this experience is extrapolated to all systems capable of distinction.
- *Operational criterion*: the hypothesis remains valid as long as no system is discovered that violates the asymmetry.

This methodological move is neither pure phenomenological description nor naive realism. It could be called *operational phenomenology*—an approach that employs the phenomenological method to access structures subsequently discovered to be universal.

The Ontological Status of the Field of Distinction: An Attempt at Formalization

The concept of "field of distinction" remains intuitively appealing but operationally vague unless we specify a way to formalize it. Within the scope of this work, we do not claim to have a finished theory, but we propose a direction in which such formalization is possible—through the apparatus of *category theory and topos theory*.

The *field of distinction* can be represented as a *topos* (a category with certain properties), where:

- *Objects* are various states of the field (forms) distinguishable at a given scale.

- *Morphisms* are scale operators $\Gamma(\lambda)$ specifying transitions between levels.
- The *subobject classifier* (truth value) arises from the structure of the field itself, not imported from outside—this corresponds to our thesis about invariants arising within distinction.

Different scales correspond to different *topoi* (or different objects within a single topos) with different internal logics. Classical arithmetic arises in that topos (or that object) where objects become stable and isolated.

This does not solve all problems—for example, the question of how the topos itself arises from the primitive "field" remains. However, such an approach:

- Allows the concept of "field of distinction" to be given formal status within existing mathematics.
- Explains why different scales may have different logics (different topoi).
- Connects our ontology with leading programs in the foundations of mathematics.

Within the scope of this work, we leave this sketch at the level of indicating a direction. Its full realization is a task for collaboration with mathematicians specializing in category theory and topos theory.

The question of "before" and "after" loses meaning, because the concept of time already presupposes the presence of an act of distinction (distinguishing "before" and "after"). To ask whether the field of distinction existed before the first observer is implicitly to introduce an observer who distinguishes that field. This is a paradox of self-reference, analogous to the question of what was before the Big Bang within GR: the question is correctly posed only from within the temporal structure, which itself arises together with the distinguished field.

We propose a third path, reducible neither to realism nor to idealism. The field of distinction is neither "objective reality" (which exists prior to any observation) nor a "construction of consciousness" (which arises together with the observer). It is an ontological primitive from which both objects and observers grow. The act of distinction is not a psychological event, but a structural relation that can be realized in quantum interaction (a particle distinguishing itself from its environment through correlations), in a sensor, and in consciousness.

In this sense, the "I" as a point of reference is not a property of human consciousness, but an ontological invariant that manifests at all levels of complexity. But it manifests as a point of reference only where there is sufficient stability of distinction for self-identity to arise. In an atom, this structure does not reach the level of complexity that could be called an "I," but the structural position (distinguishing itself from its environment) is already present in any interaction.

To the question "did the field of distinction exist before the appearance of observers?" we answer as follows:

- Within our ontology, this is a question about what existed before the very act of distinction arose, which is meaningless because the question itself already presupposes the distinction of "before" and "after," and therefore is already within the act of distinction.

- We do not assert that the observer "creates" the field. We assert that the field and the point of reference (observer in the broad sense) arise together in the act of distinction, which can be realized at different levels of complexity. At the level of quantum interactions, this act is realized as correlation; at the level of biology, as sensation; at the level of consciousness, as reflection.

- Therefore, the question of what was "before" the first conscious observer should be reformulated: how are the various regimes of distinction (from quantum to reflexive) structured, and how do they relate? This is no longer a metaphysical dispute, but an operational research program.

Classical philosophical dichotomies (realism/idealism, subject/object) arise from the implicit assumption that the "I" and the "world" are two separate objects that must somehow be connected. Our ontology asserts that they are two aspects of a single act of distinction, and therefore the question of their "primacy" is ill-posed. This is not an evasion, but a consistent application of the phenomenological stance: we begin with what is given (the act of distinction), and we do not introduce entities that are not given.

Thus, the MPO-System provides a way to move beyond the classical dilemma. Within our work, we do not choose between realism and idealism, because this alternative itself is built on assumptions we have refused to accept. Instead, we work with the field of distinction as a primitive from which both objects and points of reference, including the "I" in its various manifestations, grow. The question "was the field there before the observer?" is replaced by "how are the various levels and regimes of distinction structured, and how are they related?" This makes ontology operational and open to dialogue with physics, biology, and cognitive science.

The Human Position on the Scale: Asymmetry and "Shift Toward the Boundary of the Microworld"

The Initial Intuition

Humans occupy a special place on the scale: they are "shifted" toward the boundary of the microworld, rather than being at the center. This has important consequences for understanding the possibilities and limitations of cognition.

When moving toward smaller scales, humans quickly encounter physical limits: the atom, the electron, quantum fluctuations. There, structure disintegrates (in the sense of a change of ontology), and complex phenomena (consciousness, computation, social structures) are impossible. When moving toward larger scales, there is no such limit: one can conceive of planets, stars, galaxies, and even if the universe is finite, the very possibility of distinction is preserved. This means that humans are at the boundary of the microworld, not at the center of a symmetric scale.

Consequences for Cognition

- What we call the "microworld" is not simply "small," but the limiting regime of distinction where stable structures disappear (being replaced by other ontologies).

- Our cognitive apparatus (including language, mathematics, logic) has been formed in that range of scales where objects are stable, isolated, and countable. To transfer these tools to other ranges is to commit a category error.

- This explains why quantum mechanics seems "strange": it describes a regime where our familiar notions (object, trajectory, isolation) cease to function.

Originality

Usually, the "intermediate position" of humans is interpreted as fortunate for cognition (we see both the micro and the macro). Let us also note that this position is asymmetric: we are closer to the microworld, and this is why the quantum world seems more "alien" to us than the cosmological. This provides a new perspective on the status of fundamental theories.

Cosmological Implications

The Initial Intuition

Cosmology is not the science of "the universe as an object," but the study of regimes of distinction at cosmological scales.

How This Emerged in the Course of Analysis

- **Dark energy** is not a "substance," but an effect that arises when transitioning to a scale where the structure of distinction changes its mode. At larger scales, gravity weakens, and a new principle (repulsion) emerges, one that has no analogue in the microworld.

- **The cosmological horizon** is not "what we do not see," but the boundary of the regime of distinction: beyond it, different principles operate, possibly having no analogy at our scale. The question of what is "there" is ill-posed, because there are no conditions for distinguishing "there" from "here."

- **The Big Bang** is not "the beginning of time," but the limit of limitation itself: the point beyond which distinction is impossible. The question of what was "before" is meaningless, because there are no conditions for distinguishing "before" from "after."

- **The arrow of time** is not a consequence of entropy, but a manifestation of the asymmetry of scale: time "flows" in the direction of increasing scale, where new levels of organization open up; in the direction of decreasing scale, time "collapses" (quantum processes are reversible).

- **The fine-tuning of constants** is neither chance nor design, but the conditions for the possibility of the field of distinction in the mode we call "our universe." Other constants would entail another mode, in which distinction is either impossible or operates differently.

Phenomenological Interpretation vs. Physical Model

The proposed understanding of cosmological phenomena (dark energy, horizon, singularity, arrow of time) as effects arising from a change in the regime of

distinction at extreme scales does not claim to supplant physical theories. It does not derive the equation of state for dark energy (ω), nor does it predict the value of the cosmological constant, nor does it provide a new equation for the horizon or a mechanism for inflation. Its task is different: to indicate what kind of ontological shift could render the observed effects natural rather than paradoxical, and why attempts at direct scaling (for example, quantizing the vacuum to explain dark energy) encounter fundamental difficulties.

Physicists may (and should) continue to work with the ΛCDM model, modified theories of gravity, scalar fields, loop quantum gravity, etc. Our interpretation does not compete with these approaches, but offers them a meta-language: it explains why effects may arise at cosmological scales that have no analogue in the microworld, and why a unified theory, if it is possible, should describe not so much "laws" as the rules of transition between regimes of distinction.

The distinction is fundamental: we are not building an alternative physics, but proposing an ontological framework within which different physical models can be understood as different ways of describing the same phenomenon—the change of the regime of distinction at extreme scales. Which of these ways proves adequate is a question for empirical physics, not phenomenology.

Universality of the Framework

The approach outlined above provides a unified framework for reinterpreting key cosmological themes: the horizon, singularity, arrow of time, as well as the nature of dark energy, are understood not as isolated puzzles, but as consequences of the fundamental asymmetry of scale. These ideas do not replace physical models, but provide them with a broader context in which it becomes understandable why physics cannot give a "final answer" to questions about the boundaries of the universe: because these boundaries themselves constitute the possibility of asking questions.

Uniqueness of the Idea

Cosmology becomes not a branch of physics, but an ontological investigation of how regimes of distinction are structured at extreme scales. This resolves many paradoxes (for example, the black hole information paradox, the singularity problem) by redefining the question: not "how do the laws work?" but "how is the boundary structured, beyond which laws cease to have meaning?"

Conclusion: Findings and a Research Program

Main Findings

Asymmetry of scale is a condition for the possibility of distinction, not a consequence of physical laws. Arithmetic is a local dialect that arises only where the field of distinction stabilizes into stable objects. The "I" is not a property of consciousness, but a universal principle of distinction. Cosmology is the study of regimes of distinction at extreme scales. Operational phenomenology is a method for constructing the foundations of mathematics and physics from the immediate phenomena of distinction.

Directions for Further Work

Operationalization: transition from metric to topology (n_{min}, n_{max} as ordinal indices, graph approach).

Application to "hard problems": quantum gravity as an investigation of the boundaries of distinction; consciousness and AI—as questions about thresholds of complexity and the emergence of the point of reference.

Formalization: an axiomatic of the field, the operator $\Gamma(\lambda)$, and asymmetry; development in the direction of category theory and topos theory.

Operationalizing Asymmetry: From Metric to Topology

Introducing the parameters λ_{min} and λ_{max} allowed us to fix asymmetry as a relation of limits. However, these parameters still imply the existence of a numerical metric, which contradicts the goal of constructing a "numberless" phenomenological mathematics. Therefore, it is necessary to propose a way to assess asymmetry without using numbers, relying on topological invariants and graph properties of the field of distinction.

Asymmetry as Number of Levels of Connectivity

Instead of the metric λ, one can operate with the order of connectivity—the number of levels at which the structure preserves its topological properties (connectivity, number of components, presence of holes). By successively applying the operator $\Gamma(\lambda)$ (decreasing scale), we record:

- n_{min}—the number of steps after which the topological invariant (for example, the number of connected components) ceases to be defined or changes abruptly.
- n_{max}—the number of steps when increasing scale, after which new levels cease to arise (or their emergence is unlimited).

Asymmetry is expressed as $n_{max} \gg n_{min}$ or, in the limit, as n_{max} being infinite with n_{min} finite. This is no longer a numerical metric, but a topological index measured in steps, not in units of length.

Graph Approach: From Connectivity to Complexity

The field of distinction can be represented as a network (graph), where nodes are stable inhomogeneities and edges are the relations between them. When scale changes, the graph transforms:

- When decreasing scale (zooming in), nodes may break down into subnetworks, which in the limit lose connectivity (dissolve into isolated vertices). The number of steps until dissolution is n_{min}.
- When increasing scale (zooming out), nodes merge, forming ever larger clusters. If the process can continue indefinitely, $n_{max} \to \infty$.

In this language, asymmetry is the asymptotic behavior of the graph: in one direction—collapse to isolated vertices; in the other—unbounded growth. This allows us to do without numbers, operating only with the relation "connected / disconnected" and the order of steps.

Measuring Asymmetry as an Order of Steps

Measurement does not have to be numerical. In science, there are many ways of measuring that are not reducible to numerical scales. The ordinal scale (more/less), topological invariants (connected/disconnected, number of

components), graph metrics (distance between vertices measured in steps)—all these are forms of measurement that do not require a metric in the Euclidean sense. We propose to measure asymmetry precisely in this way: the number of steps until a limit is reached when zooming in (finite) and the absence of a limit when zooming out (infinity). This is no longer a numerical metric, but an ordinal comparison.

The parameters λ_{min} and λ_{max} are introduced as ordinal indices, not as numbers in the metric sense. They can be defined without resorting to measurement in centimeters or seconds: "apply the approximation operator until the structure ceases to be distinguishable; record the number of steps." This number of steps is already a measurement, but it does not require a pre-given metric; it is based on distinguishability, which is directly given. Asymmetry is expressed in the fact that the number of steps to the limit in one direction is finite, while in the other it is infinite (or unlimited).

Clarification: ordinal vs. cardinal numbers

It is important to distinguish two senses of "number" that may be conflated in our text. To avoid a circle (using numbers to justify a rejection of numbers), they must be explicitly separated:

1. **Cardinal numbers**—what we criticize as a secondary phenomenon. This is the result of counting stable, isolated objects. It is this arithmetic ($1 + 1 = 2$) that arises only within a certain range of scales.

2. **Ordinal numbers**—these are not the result of counting, but the result of *iteration*. "First step," "second step"—this does not presuppose the existence of stable objects to be counted. It is a fixation of the sequence of operations.

The parameters n_{min} and n_{max} introduced in our operationalization refer precisely to *ordinal* numbers. We do not "count" objects; we fix how many times the approximation operator ($\Gamma(\lambda)$) can be applied before reaching a limit. This number of steps is not a metric, but a topological index measured in iterations.

Thus, we are not using the same arithmetic we are rejecting. We are using *iteration*, which is ontologically prior to counting. Iteration (repeating the same action) can be fixed before the concept of number arises: "again," "again"—and so on until the limit. The limit is fixed not by a number, but by a qualitative change of regime (the structure ceases to be distinguishable).

Distinguishing ordinal from cardinal is key to understanding our position. We are not constructing a "numberless mathematics" in the sense of rejecting any ordering. We reject *metric* and *quantitative counting*, but retain *order* (partial order of steps, hierarchy of scales). Order is primary; quantity arises on its basis within a certain range.

Examples from Practice

- In biology: "the minimal size of an organism capable of consciousness" is not a numerical constant, but a threshold that can be established experimentally by comparing species. Measurement here is classification (consciousness / no consciousness), not a metric.

- In physics: the Planck length is not a number that can be measured with a ruler, but a boundary beyond which the concepts of metric lose meaning. We speak of it as a limit, not as a value.

- In cognitive science: the "threshold of distinction" (for example, the minimal size of an object a person can still see) is measured in minutes of arc, but this is a secondary metric imposed on the primary distinction. We work with the primary distinction, which does not require numerical fixation.

Numbers are not forbidden, but they are not the foundation. We are not against numbers in general. We are against numbers (and metrics) being considered the only language for describing reality. In our ontology, numbers arise at a certain stage—when the field of distinction stabilizes into isolated, countable units. To measure asymmetry at the fundamental level, ordinal comparisons and topological invariants suffice. If later it becomes necessary to impose a numerical metric (for example, to measure exactly how many steps to the limit), this can be done, but it will already be an interpretation within a specific system, not an ontological requirement.

Verifiability Is Preserved

The hypothesis of scale asymmetry remains verifiable because it makes predictions about what can be observed and what cannot. For example:

- One cannot discover a system whose structure, under arbitrarily deep approximation, is preserved without a change of ontology.

- One cannot discover a system whose structure, under arbitrarily far distancing, disappears (i.e., reaches a finite limit).

These predictions can be tested—in physics, in cognitive science, and in mathematical constructions. If such a system were discovered, the hypothesis would be refuted.

Measurement without numbers is not only possible, but widely used in science (ordinal scales, topological classifications). Our operationalization relies on distinguishability and the order of steps, which does not require introducing a metric before it emerges from the field of distinction itself. There is no contradiction here between "measurement" and "rejection of numbers."

Empirical Examples

- Human perception: the visual field, when zoomed in, ceases to be connected at the level of a single pixel ($n_{min} \sim$ 5–10 steps); when zoomed out, connectivity is preserved at the levels of room, house, city, landscape ($n_{max} \to \infty$).

- Quantum system: the wave function, when "zoomed in" to the Planck scale, loses its metric structure, transitioning to spin networks (n_{min} finite); when "zoomed out," it becomes more complex, transitioning to the classical world ($n_{max} \to \infty$).

Application to "Hard Problems": Caution and Prospects

The proposed framework has already been applied to consciousness, quantum gravity, and AI. However, in the case of AI and the hypothetical "consciousness of a galaxy," a strict distinction must be made between the phenomenological point of reference and the ontological principle of distinction, as well as between the empirically observable and the speculative.

Quantum Gravity: The Planck Scale as a Boundary of Distinction

The interpretation of the Planck length as a limit beyond which distinction ceases to function finds support in some approaches to quantum gravity (loop quantum gravity, noncommutative geometry). Here, the Planck scale is not simply a small length, but a boundary where spacetime loses metric meaning, and the concepts of "distance" and "trajectory" are replaced by algebraic structures (spin networks, groupoids). This agrees with our idea of a change of ontology: in the limit $\Gamma(\lambda_p)$, the operator of distinction can no longer be defined as a mapping that preserves the topology of the continuum.

Further development could consist in formalizing the transition from a discrete network to a continuum through a category of fractal or graph models, which is a subject of active research in theoretical physics, but lies beyond the scope of our phenomenological work.

Consciousness: Complexity Threshold and the Possibility of Alternative Substrates

In the section on consciousness, the thesis was advanced about the possibility of a "galactic consciousness" or the unlimited scaling of AI. This claim requires a strict qualification: it pertains to ontological possibility, not empirical reality. Consciousness (as a regime of distinction with a point of reference) requires, at a minimum, the presence of stable connections sufficient for the emergence of self-identity. Theoretically, nothing forbids such a structure from existing on a galactic scale, if there are stable channels of communication and feedback. However, empirically, we observe neither galactic consciousness nor arbitrarily scalable AI consciousness. Therefore, within the framework of operational phenomenology, we can only assert:

- In accordance with the previously described asymmetry of the scale axis, below (small scales) there is a rigid threshold: an atom does not possess an "I."
- Above (large scales), there is no fundamental ontological prohibition, but practical limitations apply (speed of light, thermodynamics, structural stability).

Thus, the thesis about the "consciousness of a galaxy" should be formulated not as a prediction, but as an indication of asymmetry: unlike the microworld, where complexity is impossible, the macroworld in principle allows it, though it is realized only under certain conditions.

Artificial Intelligence: Complexity Threshold and Emergence

Similarly, for AI, our framework allows us to pose the question not as "does it have consciousness?" but as "at what scale of complexity (number of parameters, connectivity, recursion) does a regime analogous to the point of

reference emerge?" This is an operational question that can be investigated experimentally, without speculation about "soul" or "qualia."

(Cf. Giulio Tononi's Integrated Information Theory (IIT), where consciousness is linked to the quantity Φ—a measure of a system's integrated information [Tononi, 2008; Tononi & Koch, 2015]. Unlike IIT, where consciousness is defined by a quantitative threshold, our framework emphasizes a topological transition: the emergence of the point of reference occurs when the form of distinction stabilizes at a certain scale of complexity, not upon reaching a fixed numerical value of Φ.)

Formal Phenomenology of Scale: Toward an Axiomatic and a Calculus

The proposed language (field, operator $\Gamma(\lambda)$, form, boundary) remains a set of definitions unless rules for composing operators and criteria for equivalence of forms are given. To transform it into a rigorous theory, an axiomatic is needed, even if minimal, as well as a sketch of a formal calculus that would allow operating with these concepts without reducing them to numbers.

Basic Axioms

1. *Axiom of the field*: There exists a field Φ that has no internal structure prior to the application of the operator of distinction.

2. *Axiom of the operator*: For any $\lambda \in \Lambda$ (a set of distinguishable modes), an operator $\Gamma($ $thatPhi$ into a structured field $\Phi(\lambda)$, where $\Phi(\lambda)$ contains forms—holistic configurations distinguishable at the given λ.

3. *Axiom of hierarchy*: There exists a partial order relation on Λ such that if $\lambda_1 < \lambda_2$, then $\Phi(\lambda_1)$ contains a more detailed structure than $\Phi(\lambda_2)$, and vice versa.

4. *Axiom of boundary*: For each λ, there exists a boundary $\partial(\lambda)$—a set of points (or elements) in $\Phi(\lambda)$ where the form ceases to be distinguishable. The boundary does not belong to $\Phi(\lambda)$ and determines the transition to another λ'.

5. *Axiom of asymmetry*: The set Λ has a minimal element λ_{min}, below which $\Gamma(\lambda)$ is not defined (there are no forms), and has no maximal element ($\lambda_{max} \to \infty$).

Composition Rules

- Composition of operators: $\Gamma(\lambda_2) \circ \Gamma(\lambda_1)$ is defined only if $\lambda_2 > \lambda_1$ (zooming in) or $\lambda_2 < \lambda_1$ (zooming out). Order matters: $\Gamma(\lambda_2) \circ \Gamma(\lambda_1) \neq \Gamma(\lambda_1) \circ \Gamma(\lambda_2)$ if λ_1 and λ_2 are incomparable. This non-commutativity reflects paradoxicality (analogous to PPU).

- Preservation of invariants: Some topological properties (number of connected components, presence of holes) may be preserved under composition for a certain range of λ. These are invariants of distinction.

Example of a Simplest Algebra of Distinction

As an illustration of the proposed principles, we can return to the graph approach already described. Consider the simplest field—a linear order of N distinguishable positions. The operator $\Gamma(\lambda)$ for $\lambda = k$ (integer) is the grouping of neighboring positions into blocks of size k. Then:

- When decreasing k (zooming in), the number of blocks increases until it reaches N (individual positions) $\rightarrow$ a limit.
- When increasing k (zooming out), the number of blocks decreases, tending to 1 (a single whole) $\rightarrow$ no limit (one can continue, grouping blocks into superblocks).

This is the simplest model demonstrating asymmetry: below—a limit (individual positions), above—infinity (one can enlarge indefinitely). It does not require numbers: all operations are groupings, the invariant is the number of blocks.

Sketch of a Formal Calculus for the Scale Operator

The proposed ontology of scale as an operator of distinction can be developed in the direction of a formal language that, while remaining predominantly qualitative, allows describing transformations of forms and fixing invariants. Below is a sketch of such a calculus—not a rigorous theory, but a conceptual prototype illustrating a possible path.

Basic Objects

- **Field** Φ—unstructured givenness, prior to any distinction.
- **Scale operator** $\Gamma(\lambda)$—an action that, at a given "mode" λ (not necessarily numerical), distinguishes forms from the field. λ belongs to the set of modes Λ, which has the structure of a partially ordered set (for example, "deeper" / "shallower").
- **Form** $F(\lambda) = \Gamma(\lambda)[\Phi]$—the result of applying the operator: a holistic configuration distinguishable at the given λ.

Axioms of the Calculus (in addition to the basic ones)

1. *Non-commutativity*: $\Gamma(\lambda_2) \circ \Gamma(\lambda_1) \neq \Gamma(\lambda_1) \circ \Gamma(\lambda_2)$ in general. The order of scale change matters; zooming in and zooming out are irreversible.
2. *Monotonicity*: If $\lambda_1 < \lambda_2$ (λ_1 is "deeper"), then $F(\lambda_1)$ contains more detail than $F(\lambda_2)$; there exists a relation of "finer structure" between forms.
3. *Existence of a lower limit*: There exists a minimal mode λ_{min} such that $\Gamma(\lambda_{min})$ is defined, but for any $\lambda < \lambda_{min}$ the operator is not defined (or leads to forms without stable structure).
4. *Absence of an upper limit*: For any λ, there exists $\lambda' > \lambda$, and $F(\lambda')$ contains a less detailed but still distinguishable structure.

Operations on Forms

- *Comparison of forms*: $F_1 \prec F_2$ means that F_2 is obtained from F_1 by applying $\Gamma(\lambda)$ with a larger λ (enlargement). This is a partial order.
- *Composition*: If $F_2 = \Gamma(\lambda_2)[F_1]$ and $F_3 = \Gamma(\lambda_3)[F_2]$, then $F_3 = \Gamma(\lambda_2 \cdot \lambda_3)[F_1]$ (composition of operators). "Multiplication" here is not numerical, but denotes sequential application.
- *Invariants*: A topological property $I(F)$ (for example, connectivity, number of components) is called an invariant of distinction if $I(F(\lambda_1)) = I(F(\lambda_2))$ for all λ in some range.

Simplest Implementation: A Graph Algebra of Distinction

As an illustration of the proposed principles, we can return to the graph approach already described. Let Φ be an unstructured set of "points" (or vertices).

The operator $\Gamma(k)$ (k — a natural number denoting the grouping level) acts as follows:

- The initial form $F(0)$—each point is isolated.
- $\Gamma(1)$ groups points into clusters according to the rule "neighboring points in the original graph."
- $\Gamma(2)$ groups clusters into larger blocks, and so on.

Then:

- When decreasing k (moving from k to $k-1$), we return to a finer structure—this is zooming in.
- When increasing k (enlarging), we obtain more abstract forms.

Invariants: the number of connected components at level k (c_k).
Asymmetry manifests in that:

- c_k as $k \to \infty$ tends to 1 (all points merge into one cluster).
- As $k \to 0$ (i.e., when decreasing scale), the structure hits discrete points—a limit beyond which there are no new levels.

Here k is not a numerical metric, but an ordinal number of steps, and asymmetry is expressed in the fact that movement in one direction (toward increasing k) can continue indefinitely, while in the other it is finite.

Summary: The graph algebra shows how scale asymmetry manifests in topology: merging into one cluster (infinity above) and dissolution into discrete points (limit below).

Possible Extensions

- Transition to hypergraphs or topological spaces to model more complex forms.
- Introduction of a categorical structure where objects are forms and morphisms are scale operators, with emphasis on limit and colimit constructions.

- Formalization of the change of ontology as a change of category (topos) in which forms are defined.

Status of the Sketch

The proposed language is no more than an intellectual improvisation, intended to show that a formal calculus for the scale operator is possible. It does not claim mathematical rigor, but indicates a direction in which the ontology of distinction can be developed without reducing it to numbers. Further development will require collaboration with mathematicians working in the areas of category theory, topos theory, and algebraic topology, but even now it is possible to see how transformations of forms can be described and invariants fixed without recourse to a metric.

Critical Remark on the Status of Concepts

The concepts of "scale operator" and "field of distinction" may raise for a rigorous critic the concern that we remain in the realm of poetic metaphysics, never reaching formal rigor. There are two responses to this.

First, these concepts are introduced not through metaphors, but through operations that can be performed (at least mentally) by any observer: changing the scale of attention, fixing the boundary, extracting structure from the background. This is the classic phenomenological move—to begin with what is given in experience and describe it without resorting to ready-made mathematical languages.

Second, the proposed axioms and composition rules provide a first sketch of a formal language. It does not claim completeness, but shows that behind the operational concepts lies the possibility of constructing a calculus. This distinguishes our work from purely literary speculation.

Next Steps

The development of this axiomatic can proceed in the direction of:

- Generalizing to graphs (instead of a linear order) and topological spaces.
- Introducing morphisms between different $\Phi(\lambda)$ to formalize "transition" between scales.
- Connecting with existing mathematical structures (categories, topoi) as implementations of this axiomatic.

This is already a task for formal mathematics, beyond the scope of phenomenological investigation, but our framework provides an ontological foundation for it.

Terminological Glossary

Term	Meaning
Field of distinction	Ontological primitive, givenness that becomes structured in the act of distinction. Has no internal structure prior to the application of the scale operator.
Point of reference	Structural position relative to which the field of distinction is structured. In the phenomenological sense—the "I"; in the ontological sense—a universal principle present in any act of distinction.
Act of distinction	The event in which the field becomes structured and, simultaneously, the point of reference is fixed. Primary with respect to objects and observers.
Scale operator $\Gamma(\lambda)$	An operation that, at a given mode λ (depth of distinction), transforms the field into a structured forms.
Asymmetry of scale	A fundamental property of the operator of distinction: a rigid limit when moving "inward" ($\lambda \to \lambda_{min}$), absence of a limit when moving "outward" ($\lambda \to \infty$).
Form	A holistic configuration of the field, given prior to dissolution into objects and background. Changes with a change of scale as a whole, not as a sum of parts.

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This study was conducted using the conceptual apparatus of the Ontology Lab (MPO System), where the properties of Superreality and axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$) served as implicit coordinates for navigating the topic. In the final text, we consciously avoid this terminology so that the reader may focus on the heart of the matter. For those interested, a comprehensive description of the method is available in public sources.